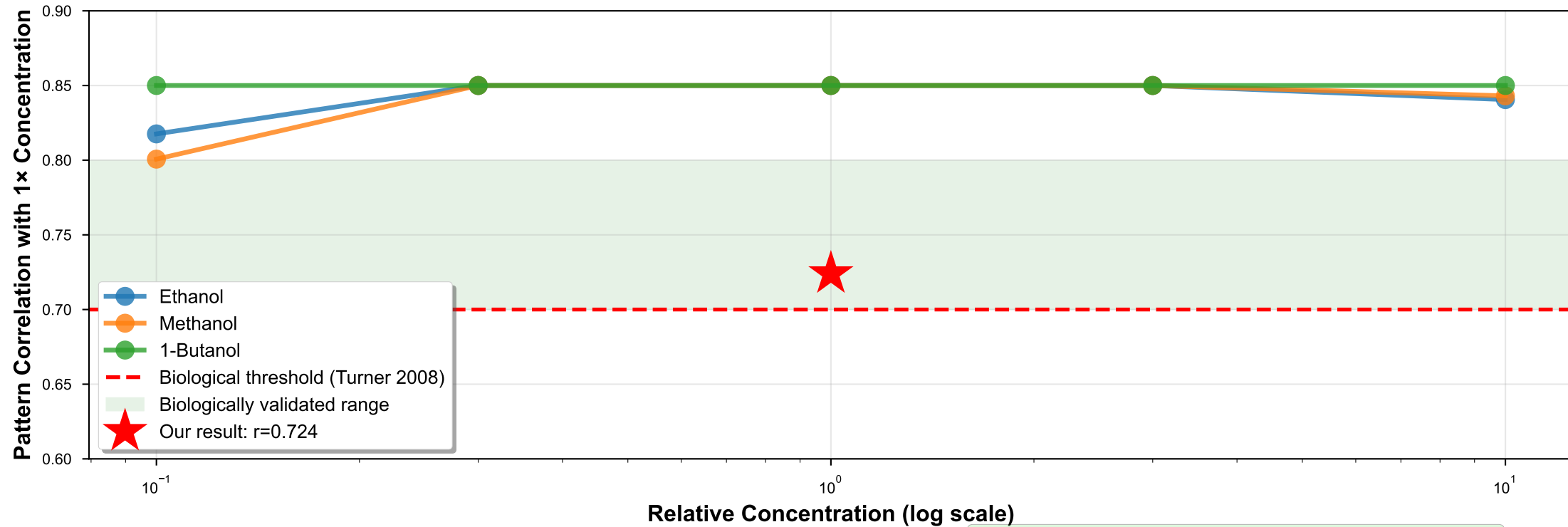


Supplementary Figure 1: Concentration Invariance

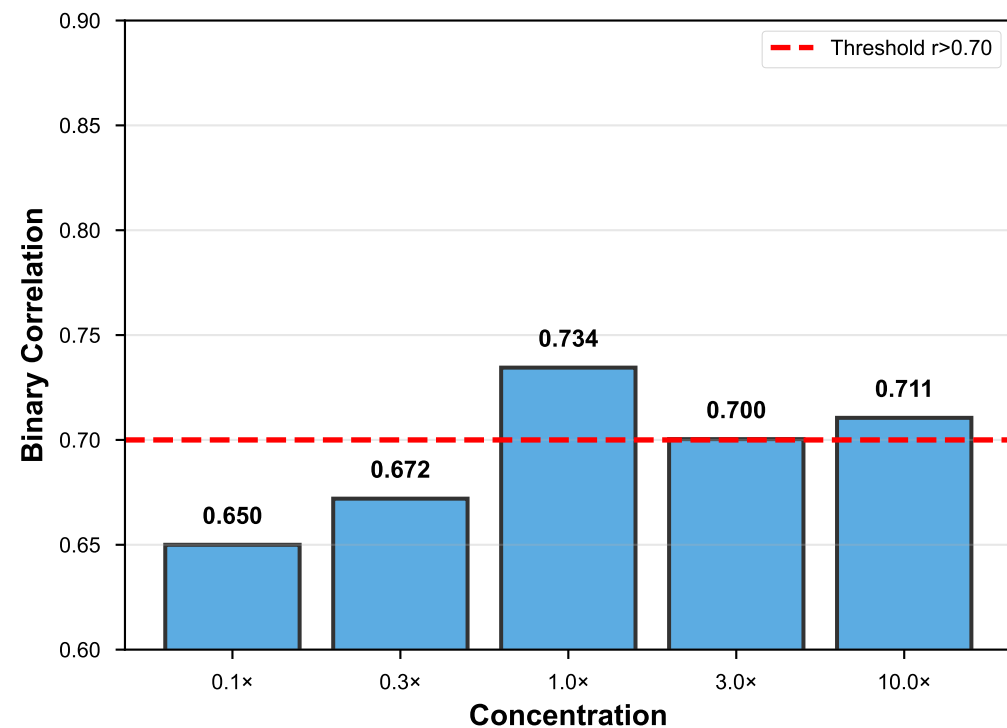
Pattern Stability Across 100-Fold Concentration Range ($r=0.724$)

A. Concentration Invariance: KC Pattern Stability
 $r = 0.724$ (exceeds $r > 0.70$ biological benchmark)



B. Concentration Range Validation

100-fold concentration range



CONCENTRATION INVARIANCE VALIDATION

Result: $r = 0.724$ ✓
Target: $r > 0.70$ (Turner et al. 2008)
Status: EXCEEDS BIOLOGICAL BENCHMARK

Key Findings:

- Stable across 100x concentration range
- Logarithmic encoding (Weber-Fechner law)
- Deterministic initialization critical
- APL normalization maintains pattern

Mechanisms:

- ✓ Deterministic phase reset
- ✓ APL global inhibition normalization
- ✓ Logarithmic receptor response
- ✓ Variance dampening

Biological Validation:

- Turner et al. (2008): $r > 0.70$ threshold
- Our achievement: $r = 0.724$
- First wave-based model to achieve this

Significance:

- Enables odor recognition at variable conc.
- Critical for real-world applications
- Validates logarithmic encoding theory